

```
Windows PowerShell
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> ruff check .
All checks passed!
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> python -m pytest -q
...
3 passed in 30.34s
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> docker build -t w12d2-ml-api:latest .
[+] Building 5.5s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 264B
=> [internal] load metadata for docker.io/library/python:3.12-slim
=> [internal] load .dockerignore
=> => transferring context: 147B
=> [internal] load build context
=> => transferring context: 3.65kB
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:f77ac9e44ae96ef2c90b8053ea08c31f8be030f824196b0ae4db6d462c84e51f
=> => resolve docker.io/library/python:3.12-slim@sha256:f77ac9e44ae96ef2c90b8053ea08c31f8be030f824196b0ae4db6d462c84e51f
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY app ./app
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:8b65dcf016917af38b7b55c523c10b813da031cad2d7e0c0060908bb83a32ad9
=> => exporting config sha256:2ec03248619f8de635e93e434df89ecda5a338301f007755902332e070c7a502
=> => exporting attestation manifest sha256:462ffb695e2de91a01b34f93c7a46bd50c58babde11089ae4099f6b15d328d9d
=> => exporting manifest list sha256:6bbb1d5f399d304f8e66f172a9918b481ff5a6e077eaa403a7349eda646ee0ac
=> => naming to docker.io/library/w12d2-ml-api:latest
=> => unpacking to docker.io/library/w12d2-ml-api:latest
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> ^C
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> docker images

INFO In Use

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
mlops-ml-api:latest                 068632f888ea                      627MB        147MB         U
production-rag-api:latest           ca25408bcb55                      310MB         75.8MB        U
w12d2-ml-api:latest                 6bbb1d5f399d                      604MB         140MB
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> docker ps

CONTAINER ID  IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
```

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CONTAINER ID  IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
ff6575a6813c  33ef7e3ff65b  "uvicorn app.main:ap..."  2 hours ago    Up 2 hours    0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp  w12d2-ml-api-container
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> docker ps

CONTAINER ID  IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D2_Docker_ML_App> |
```

W12D2: Containerising ML App x Docker token setup x Iris ML Prediction API - Swagger x + Ask Gemini

localhost:8000/docs

Iris ML Prediction API 0.1.0 OAS 3.1

/openapi.json

default

- GET** / Root
- GET** /health Health
- POST** /predict Predict

Schemas

- HTTPValidationError** > Expand all object
- IrisInput** > Expand all object
- ValidationError** > Expand all object

28° W12D2: Containerising ML App x Docker token setup x Iris ML Prediction API - Swagger x + Ask Gemini ENG IN 18:39 28-09-2026

W12D2: Containerising ML App x Docker token setup x Iris ML Prediction API - Swagger x + Ask Gemini

localhost:8000/docs#default/root_get

Iris ML Prediction API 0.1.0 OAS 3.1

/openapi.json

default

GET / Root

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X GET \
  http://localhost:8000/ \
  -H 'accept: */*'
```

Request URL

http://localhost:8000/

Server response

Code Details

200

Response body

```
{
  "message": "Iris ML API is running"
}
```

Response header

```
Content-Length: 36
Content-Type: application/json
Date: Mon, 28 Sep 2026 17:58:25 GMT
Server: gunicorn
```

Response

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Example Value - Schema

```
"string"
```

28° W12D2: Containerising ML App x Docker token setup x Iris ML Prediction API - Swagger x + Ask Gemini ENG IN 18:40 28-09-2026

W12D2: Containerising ML App x Docker token setup x Iris ML Prediction API - Swagger x + Ask Gemini

localhost:8000/docs#/default/health_health_get

GET /health/health

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X GET \
  http://localhost:8000/health/health \
  -H 'accept: */*'
```

Request URL

http://localhost:8000/health/health

Server response

Code Details

200

Response body

```
{
  "status": "healthy"
}
```

Response header

```
content-length: 20
content-type: application/json
date: Mon, 28 Sep 2026 13:11:11 GMT
server: uvicorn
```

Response

Code Description Links

200 Successful Response No links

Media type: application/json

Controls Accept header

Example Value | Schema

```
{
  "status": "healthy"
}
```

Schemas

HTTPValidationError Expand all object

W12D2: Containerising ML App x Docker token setup x Iris ML Prediction API - Swagger x + Ask Gemini

localhost:8000/docs#/default/predict_predict_post

POST /predict/predict

Parameters

No parameters

Request body `application/json`

Edit Value | Schema

```
{
  "sepal_length": 5.1,
  "sepal_width": 3.5,
  "petal_length": 5.4,
  "petal_width": 4.2
}
```

Execute Clear

Responses

Curl

```
curl -X POST \
  http://localhost:8000/predict \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "sepal_length": 5.1,
    "sepal_width": 3.5,
    "petal_length": 5.4,
    "petal_width": 4.2
  }'
```

Request URL

http://localhost:8000/predict

Server response

Code Details

200

Response body

```
{
  "prediction": 0,
  "class_name": "setosa"
}
```

Response header

```
content-length: 36
content-type: application/json
date: Mon, 28 Sep 2026 13:11:31 GMT
server: uvicorn
```

Response

Code Description Links

200 Successful Response No links